AP PRECALCULUS · TOPIC 1.8

Rational Functions and Zeros

Section 1 — Multiple Choice
Circle the letter of the correct answer

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

1 Let $r(x) = \dfrac{(x-3)(x+4)}{(x+1)(x-2)}$. What are all of the real zeros of r?

A. $-4,\ 3$ **B.** $-3,\ 4$

C. $-4,\ -1,\ 2,\ 3$ **D.** $-1,\ 2$

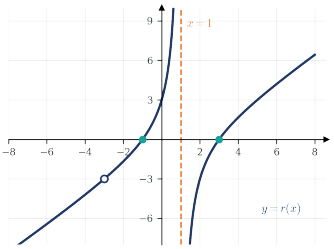
2 What are all of the real zeros of $f(x) = \dfrac{x^2-9}{x^2-x-6}$?

A. $-3,\ 3$ **B.** -3

C. 3 **D.** -2

3 The graph of the rational function r is shown. It has x-intercepts at $x=-1$ and $x=3$, a vertical asymptote at $x=1$, and a hole at $(-3,-3)$. Which is the solution set of $r(x) \le 0$?

A. $(-\infty,-1]\cup(1,3]$
B. $(-\infty,-3)\cup(-3,-1]\cup[1,3]$
C. $(-\infty,-3)\cup(-3,-1]\cup(1,3]$
D. $[-1,1)\cup[3,\infty)$



4 Which list gives all of the real zeros of $g(x) = \dfrac{x^3-x}{x^2+4}$?

A. $-1,\ 1$ **B.** There are no real zeros

C. $-2,\ 2$ **D.** $-1,\ 0,\ 1$

5 What is the real zero of $r(x) = \dfrac{1}{x-2} - \dfrac{3}{x+4}$?

A. -7 **B.** 5

C. -5 **D.** r has no real zeros

6 Which is the solution set of $\dfrac{x+2}{x-3} \le 0$?

A. $[-2,3]$ **B.** $(-\infty,-2]\cup(3,\infty)$

C. $(-2,3)$ **D.** $[-2,3)$

7 Which function has exactly one real zero?

A. $f(x) = \dfrac{x^2-4}{x-2}$ **B.** $f(x) = \dfrac{x^2-4}{x^2+1}$

C. $f(x) = \dfrac{(x-2)(x+1)}{x+3}$ **D.** $f(x) = \dfrac{x^2+4}{x-1}$

8 Let $f(x) = \dfrac{(x-4)^2(x+1)}{(x-4)(x+2)}$. Which statement is true?

A. f has zeros at $x=-1$ and $x=4$
B. f has a zero at $x=4$ only
C. f has exactly one zero, $x=-1$, and a hole at $(4,0)$
D. f has zeros at $x=-2$ and $x=-1$

AP PRECALCULUS · TOPIC 1.8 · SECTION 2

Rational Functions and Zeros

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

Question 9 Calculator allowed · 6 points

A biologist models the rate of change of the fish population in a lake, in hundreds of fish per month, by $R(t) = \dfrac{2(t-6)(t+1)}{t+3}$, where $t \geq 0$ is the time in months since the study began.

(a) Find the zero of R in the context of the problem. Explain why the other zero of the numerator is not used. DOK3 2 pt

(b) Solve $R(t) \geq 0$ for $t \geq 0$ and interpret your answer in context. DOK3 2 pt

(c) Find the time at which the rate of change is exactly 4 hundred fish per month. Give your answer to three decimal places. DOK3 2 pt

Question 10 No calculator · 6 points

Let $r(x) = \dfrac{x^3 - 4x^2 - 5x}{x^2 - 2x - 15}$.

(a) Factor the numerator and the denominator, and state the real zeros of r. DOK3 2 pt

(b) Find the coordinates of the hole in the graph of r. DOK3 2 pt

(c) Solve $r(x) \geq 0$. Write your answer in interval notation. DOK3 2 pt